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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

A PROJECT REPORT

ON

"5-Question MCQ Quiz Webpage for Students"

A project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
AY 2025-26

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CERTIFICATE

This is to certify that the Project work entitled "5-Question MCQ Quiz Webpage for Students" is carried out by the **Divya Choudhary** in partial fulfillment for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, Symbiosis International (Deemed University), Pune during the academic year 2025-2026.

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DECLARATION

I hereby declare that the project titled "5-Question MCQ Quiz Webpage for Students" submitted to Symbiosis Institute of Technology, Constituent of Symbiosis International (Deemed University) Pune for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a result of original research carried out by me. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of any degree or diploma.

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ABSTRACT

This project presents the development of a 5-Question MCQ Quiz Webpage designed using HTML, CSS, and . The application provides an interactive platform where a teacher can configure five multiple-choice questions and students can attempt the quiz and receive instant feedback based on their performance. The system is designed to enhance classroom learning through a simple, engaging, and user-friendly interface.

The application includes a clean question-and-option layout, where students select one answer per question from four choices. Upon clicking the Submit button, the system evaluates all responses simultaneously and calculates the score instantly. Based on performance, feedback is displayed such as "Perfect Score!" or "Keep Practicing!", improving student engagement and motivation.

The system is entirely frontend-based and requires no backend or database. All quiz questions and correct answers are stored as arrays within the source code, making it easy for a teacher to update content by modifying a single file. The interface is designed using CSS with modern styling techniques such as gradients, card-based layout, and responsive design to ensure compatibility across desktops, tablets, and mobile devices.

Overall, the project demonstrates key concepts of frontend development including event handling, DOM manipulation, array-based data management, and responsive design. It serves as both an educational tool and a practical implementation of a web-based quiz system for classroom use.

Keywords: MCQ Quiz, Web Application, HTML5, CSS3, Score Evaluation, Frontend Development, Classroom Assessment

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CHAPTER 1: BACKGROUND AND TECHNICAL OVERVIEW

1.1 Introduction

In the modern digital era, online learning platforms have become increasingly important. Quiz-based assessment systems are widely used in education to evaluate student knowledge efficiently and provide instant feedback without manual evaluation overhead.

This project focuses on developing a 5-Question MCQ Quiz Webpage that a teacher can configure for students. The system is designed using frontend technologies such as HTML, CSS, and . The teacher embeds five multiple-choice questions directly in the source code, and students access the page via any browser to attempt the quiz.

The application provides a structured and interactive way of conducting quizzes. It displays one question at a time or all five questions on a single page, allows students to select one answer per question, and calculates the final score instantly on submission. The system ensures simplicity, accessibility, and ease of use for both teachers and students.

The main goal of this project is to demonstrate how interactive web-based assessment tools can be built using only basic frontend technologies without requiring any backend integration, database, or external API.

1.2 Core Technologies

HTML5 structures the quiz layout including question containers, radio button options, and the submit button. CSS3 handles visual styling including gradients, card shadows, and responsive layout. (ES6) stores question data as arrays of objects and drives all quiz logic — rendering questions, capturing selections, evaluating answers, and displaying the final score.

CHAPTER 2: PROBLEM STATEMENT AND MOTIVATION

2.1 Problem Statement

Traditional paper-based quizzes require manual checking, are time-consuming for teachers, and provide delayed feedback to students. Many existing digital quiz platforms are either too complex, require account creation, or depend on internet connectivity and subscription plans.

There is a need for a simple, lightweight, and teacher-configurable quiz system that can:

- **Work without any backend, server, or API**
- **Provide instant score evaluation on submission**
- **Be easily set up by a teacher with basic HTML knowledge**
- **Run on any device with a browser, without installation**

2.2 Motivation

The motivation behind this project is to create a user-friendly quiz application that enhances classroom learning and engagement. The project aims to:

- **Simplify quiz creation and score evaluation for teachers**
- **Provide students with instant feedback on their performance**
- **Demonstrate core frontend programming concepts practically**
- **Serve as a reusable template that teachers can customize per lesson**

CHAPTER 3: NOVELTY AND INNOVATIVE CONTRIBUTIONS

3.1 Novel Features

1. **Teacher-Configurable Question Bank — All 5 questions, options, and correct answers are stored in a single array. A teacher can update the quiz content by editing one file with no coding expertise required beyond basic text editing.**

2. **Instant Score Calculation — The system evaluates all five student responses simultaneously on submission and displays the score (e.g., "You scored 4 out of 5") along with a personalized feedback message.**

3. **Answer Highlight Feedback — After submission, correct answers are highlighted in green and incorrect selections are shown in red, giving students clear visual feedback on each question.**

4. **Restart Functionality — A Restart Quiz button resets all selections and re-renders the quiz, allowing students to reattempt without refreshing the page.**

3.2 Innovative Aspects

The project combines simplicity with interactivity, showing how powerful classroom tools can be built without backend technologies. It focuses on user experience with real-time feedback and clean UI design. The data-driven approach — where a single array drives the entire quiz — mirrors how modern frontend frameworks like React and Vue operate, making it an excellent learning reference.

CHAPTER 4: TECHNICAL ADVANTAGES AND PRACTICAL USEFULNESS

4.1 Technical Advantages

- **Lightweight system — no backend, database, or server required**
- **Fast performance — all logic executes instantly in the browser**
- **Easy to understand and modify — ideal for teachers with basic HTML knowledge**
- **Responsive design — compatible with desktops, tablets, and mobile phones**
- **No CORS or network issues — entirely client-side, works offline**

4.2 Practical Usefulness

- **Classroom quizzes and short formative assessments**
- **Practice tests for students before exams**
- **Online learning module checkpoints**
- **Academic project demonstration of frontend web development**

4.3 Comparison with Existing Quiz Tools

Feature	This Project	Google Forms	Kahoot!	Typeform
No Login Required	✅	❌	❌	❌
Works Offline	✅	❌	❌	❌
Instant Score	✅	✅	✅	✅
No Subscription	✅	✅	⚠	⚠
Teacher Customizable	✅	✅	✅	✅

CHAPTER 5: DETAILED METHODOLOGY / SYSTEM ARCHITECTURE

5.1 System Architecture

The system consists of:

5. **User Interface (HTML + CSS) — Quiz card layout with question text, four radio-button options per question, and a Submit button**
6. **Data Layer (Array) — All 5 questions, options, and correct answers stored as a JS array of objects**
7. **Logic Layer ( ) — Renders questions dynamically, captures student selections, evaluates answers, and displays the result**

5.2 Working Methodology

Step-by-step working:

8. **Teacher opens index.html in a browser and shares the file or URL with students**
9. **Page loads and renders all 5 questions from the data array into the DOM**
10. **Student reads each question and selects one answer from the four radio button options**
11. **Student clicks the Submit button**
12. **loops through all 5 questions and compares selected answers with correct answers**

13. **Score is calculated (0–5) and result message is displayed**
14. **Correct answers highlighted green, wrong answers highlighted red for review**
15. **Student clicks Restart to reset and reattempt the quiz**

5.3 Algorithm

- **Define questions array with 5 objects (question, options[4], answer)**
- **On page load, loop through array and render question cards into DOM**
- **On Submit click, loop through questions and check selected radio value against answer**
- **Increment score counter for each correct match**
- **Display result: score, feedback message, and colour-coded answer review**
- **On Restart click, reset all radio buttons and re-render the quiz**

5.4 File Structure

- **index.html — Main quiz page with container for question cards and submit button**
- **style.css — All styling including card layout, colours, and responsive media queries**
- **script.js — Questions data array and all quiz logic (render, evaluate, reset)**
- **README.txt — Instructions for teachers on how to customize questions**

CHAPTER 6: PRIOR ART AND RELATED WORK (Literature Survey)

6.1 Existing Systems

- **Online quiz platforms — Google Forms, Kahoot!, Quizizz, Typeform**
- **Learning management systems — Moodle, Canvas, Blackboard with built-in quiz modules**
- **Mobile quiz apps — Quizlet, Brainscape, Anki for flashcard-style assessment**

6.2 Limitations of Existing Systems

- **Require backend servers, databases, and internet connectivity**
- **Complex setup involving account creation, login, and subscription management**
- **Subscription costs or rate-limited free tiers for advanced features**
- **Limited customization without technical knowledge of the platform**
- **Not suitable for quick classroom use without prior platform setup**

6.3 How This Project Addresses the Gap

The 5-Question MCQ Quiz Webpage removes every barrier present in existing systems. There is no account, no login, no internet required, and no subscription. A teacher edits a single file to set questions and answers, then shares the HTML file with students directly. This makes it the fastest and most accessible way to conduct a digital quiz in a classroom setting.

CHAPTER 7: APPLICATIONS AND DEPLOYMENT AREAS

7.1 Applications

- **Schools and colleges — for formative assessments and class tests**
- **Online exams — as a lightweight digital substitute for paper-based tests**
- **Competitive exam preparation — for topic-wise practice sets**
- **Self-learning platforms — as checkpoint quizzes within e-learning modules**
- **Academic project demonstrations — showcasing frontend development skills**

7.2 Deployment

- **GitHub Pages — free static hosting, shareable via URL with no server setup**
- **Local browser — teacher shares the index.html file directly, no internet needed**
- **Educational labs — deployed on lab intranet for whole-class access**

- **Netlify or Vercel — one-click deployment for permanent online access**

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The 5-Question MCQ Quiz Webpage successfully demonstrates an interactive and efficient quiz system using only frontend technologies. It provides teachers with a customizable, zero-setup tool to conduct student assessments, and gives students instant score evaluation with clear answer feedback. The project is lightweight, responsive, and easy to deploy, making it highly practical for real classroom use.

8.2 Future Scope

- **Add database support to store student scores and quiz history**
- **Implement a user login system to track individual student performance**
- **Allow teachers to add more questions dynamically through a form interface**
- **Add a countdown timer for time-limited quiz scenarios**
- **Build a leaderboard system to display top scores across the class**
- **Develop a mobile app version using React Native or Flutter**
- **Enable question randomization to prevent answer sharing between students**

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APPENDICES

Appendix A: Source Code

The source code of the project is implemented using HTML, CSS, and . The application includes question rendering logic, answer evaluation, score calculation, and restart functionality.

```
quiz-project/
│
├── index.html     →  Quiz page with question containers & submit button
├── style.css      →  Styling, card layout, colour feedback, responsive design
├── script.js      →  Questions array, render logic, score evaluation, restart
└── README.txt     →  Teacher instructions for customizing questions
```

Appendix B: System Requirements

Component	Requirement
Browser	Chrome / Firefox / Edge (latest version)
Internet Connection	Not required (fully offline)
Backend / Server	Not required
Programming Language	HTML5, CSS3, (ES6+)

Operating System	Windows / macOS / Linux / Android / iOS

Appendix C: Glossary of Terms

Term	Definition
HTML	HyperText Markup Language — standard language for structuring web pages.
CSS	Cascading Style Sheets — defines visual presentation and layout of HTML elements.
	A scripting language that enables dynamic, interactive behaviour in web pages.
MCQ	Multiple Choice Question — a question with one correct answer among several options.
DOM	Document Object Model — a programming interface for HTML documents.